

Border Discontinuity Part 2

Setting up the groups importing packages, and cleaning names

```
set.seed(123)
library(blavaan)
```

Loading required package: Rcpp

This is blavaan 0.6-1

On multicore systems, we suggest use of `future::plan("multicore")` or `future::plan("multisession")` for faster post-MCMC computations.

```
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

✓ dplyr	1.2.1	✓ readr	2.2.0
✓ forcats	1.0.1	✓ stringr	1.6.0
✓ ggplot2	4.0.3	✓ tibble	3.3.1
✓ lubridate	1.9.5	✓ tidyr	1.3.2
✓ purrr	1.2.2		

— Conflicts — tidyverse_conflicts() —

✗ dplyr::filter() masks stats::filter()

✗ dplyr::lag() masks stats::lag()

ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

```
library(readr)
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(bayesplot)
```

This is bayesplot version 1.16.0

- Online documentation and vignettes at mc-stan.org/bayesplot
- bayesplot theme set to `bayesplot::theme_default()`
 - * Does `_not_` affect other ggplot2 plots
 - * See `?bayesplot_theme_set` for details on theme setting

```
library(bayestestR)
high_peak_areas <- c(
  "New Mills West Ward",
  "New Mills East Ward",
```

```

    "Buxton Central Ward",
    "Dinting Ward",
    "Old Glossop Ward",
    "Hadfield South Ward",
    "Hadfield North Ward"
  )
gmppanel_prop = read_csv("c:/Users/User/Downloads/gmppanel_prop.csv")

```

Rows: 264 Columns: 6

— Column specification —

Delimiter: ","

chr (3): Location, Area, Classification

dbl (3): Year, Count, Proportion

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```

widepanel <- pivot_wider(
  gmppanel_prop,
  id_cols = c(Location, Year),
  names_from = Classification,
  values_from = Proportion,
  values_fn = first
)
widepanel = widepanel %>% filter(Location != c("Tameside District", "High Peak District"), Year
  mutate(highpeak = ifelse(Location %in% high_peak_areas, 0, 1),
         established = ifelse(Year >= 2011, 1, 0),
         gmp = established * highpeak)

widepanel_clean = widepanel %>% clean_names()

```

Defining the pooled structural model with the SUR residual covariance and treatment coefficients note that the covariances absorb both between and within variation

```

sur_model_clean <- '
# Outcome Regressions within towns
higher_managerial_administrative_professional_occupations ~ bhm*gmp + established+highpeak
intermediate_occupations ~ bim*gmp + established+highpeak
routine_manual_occupations ~ br*gmp + established+highpeak

# SUR Residual Covariances
higher_managerial_administrative_professional_occupations ~~ phim*intermediate_occupations
higher_managerial_administrative_professional_occupations ~~ phr*routine_manual_occupations
intermediate_occupations ~~ pimr*routine_manual_occupations

#defined parameters including implied effect on unemployment
gap_routine_manage := br-bhm
gap_manual_intermediate := br-bim
gap_highm_intermediate := bhm-bim
polarization := (bhm + br) - bim
implied_ch_in_unemployment := -(bhm+bim+br)

```

```
'
priorbetas = dpriors(nu = "normal(0,10)",beta = "normal(0,0.5)")
```

Also define a prior for coefficients and intercept in general we use a loose normal(0,0.5) prior with 95 percent of expected parameter values representing proportion changes lying between ± 0.98 . Now estimate the model by MCMC, using 8 chains/cores for speed

```
fitbayes <- bsem(
  sur_model_clean,
  dp = priorbetas,
  data      = widepanel_clean,
  n.chains = 8,
  cores = 8)
```

Computing post-estimation metrics (including lvs if requested)...

The pooling model uses both within and between variation for estimating coefficients which means it can be somewhat oblivious to the unique time invariant characteristics that each town on either side of the border might have historically which might get mixed up with the cross border effect. Therefore a two way within transformation is used where we first group by town and subtract out the means and then group by year and subtract out the means. This means that coefficients now represent the average of within town deviations from the mean associated with joining greater manchester effectively using each towns variation overtime as a control for itself. Subtracting out the year means similarly controls for any shocks hitting all towns at the same time

```
widepanel_demeaned <- widepanel %>%
clean_names() %>%
group_by(location) %>%
mutate(across(where(is.numeric), ~ .x - mean(.x, na.rm = TRUE))) %>%
group_by(year) %>%
mutate(across(where(is.numeric), ~ .x - mean(.x, na.rm = TRUE))) %>%
ungroup()
sur_model_within <- '
# Outcome Regressions within towns
higher_managerial_administrative_professional_occupations ~ bhm*gmp
intermediate_occupations ~ bim*gmp
routine_manual_occupations ~ br*gmp

# SUR Residual Covariances
higher_managerial_administrative_professional_occupations ~~ phim*intermediate_occupations
higher_managerial_administrative_professional_occupations ~~ phr*routine_manual_occupations
intermediate_occupations ~~ pimr*routine_manual_occupations

#defined parameters including implied effect on unemployment
gap_routine_manage := br-bhm
gap_manual_intermediate := br-bim
gap_highm_intermediate := bhm-bim
polarization := (bhm + br) - bim
implied_ch_in_unemployment := -(bhm+bim+br)'
```

We now estimate the within town fixed effects model unbiased by time invariant unobservables:

```
fitbayeswithin <- bsem(
  sur_model_within,
  dp = priorbetas,
  data      = widepanel_demeaned,
  n.chains = 8,
  cores     = 8
)
```

Computing post-estimation metrics (including lvs if requested)...

```
summary(fitbayeswithin)
```

blavaan 0.6.1 ended normally after 1000 iterations

Estimator	BAYES
Optimization method	MCMC
Number of model parameters	9
Number of observations	54
Statistic	MargLogLik
Value	328.105
	PPP
	0.486

Parameter Estimates:

Regressions:

	Estimate	Post.SD
higher_managerial_administrative_professional_occupations ~		
gmp (bhm)	-0.063	0.036
intermediate_occupations ~		
gmp (bim)	0.065	0.028
routine_manual_occupations ~		
gmp (br)	0.018	0.024
pi.lower pi.upper Rhat Prior		
-0.133 0.006 1.000 normal(0,0.5)		
0.009 0.120 1.000 normal(0,0.5)		
-0.030 0.065 1.000 normal(0,0.5)		

Covariances:

	Estimate
.higher_managerial_administrative_professional_occupations ~~	
.intrmd_ (phim)	-0.001
.rtn_mn_ (phr)	-0.001
.intermediate_occupations ~~	
.rtn_mn_ (pimr)	-0.000
Post.SD pi.lower pi.upper Rhat Prior	

0.000	-0.000	1.001	lkj_corr(1)
0.000	-0.000	1.000	lkj_corr(1)
0.000	0.000	1.000	lkj_corr(1)

Variances:

	Estimate	Post.SD	pi.lower	pi.upper	Rhat	Prior
.hghr_mngrl_d__	0.002	0.000	0.001		1.000	gamma(1,.5)[sd]
.intrmdt_ccptns	0.001	0.000			1.001	gamma(1,.5)[sd]
.rtn_mnl_ccptns	0.001	0.000	0.000		1.000	gamma(1,.5)[sd]

Defined Parameters:

	Estimate	Post.SD	pi.lower	pi.upper	Rhat	Prior
gap_routin_mng	0.081	0.054	-0.023	0.187	1.000	
gap_mnl_ntrmdt	-0.047	0.038	-0.122	0.027	1.000	
gp_hghm_ntrmdt	-0.129	0.059	-0.242	-0.014	1.001	
polarization	-0.111	0.055	-0.218	-0.000	1.000	
impld_ch_n_nmp	-0.020	0.015	-0.050	0.010	1.001	

```
blavCompare(fitbayes,fitbayeswithin)
```

Warning:

11 (20.4%) p_waic estimates greater than 0.4. We recommend trying loo instead.

Warning:

6 (11.1%) p_waic estimates greater than 0.4. We recommend trying loo instead.

WAIC estimates:

```
object1: -478.395
object2: -760.017
```

ELPD difference & SE:

```
-140.811 12.665
```

LOO estimates:

```
object1: -477.879
object2: -759.596
```

ELPD difference & SE:

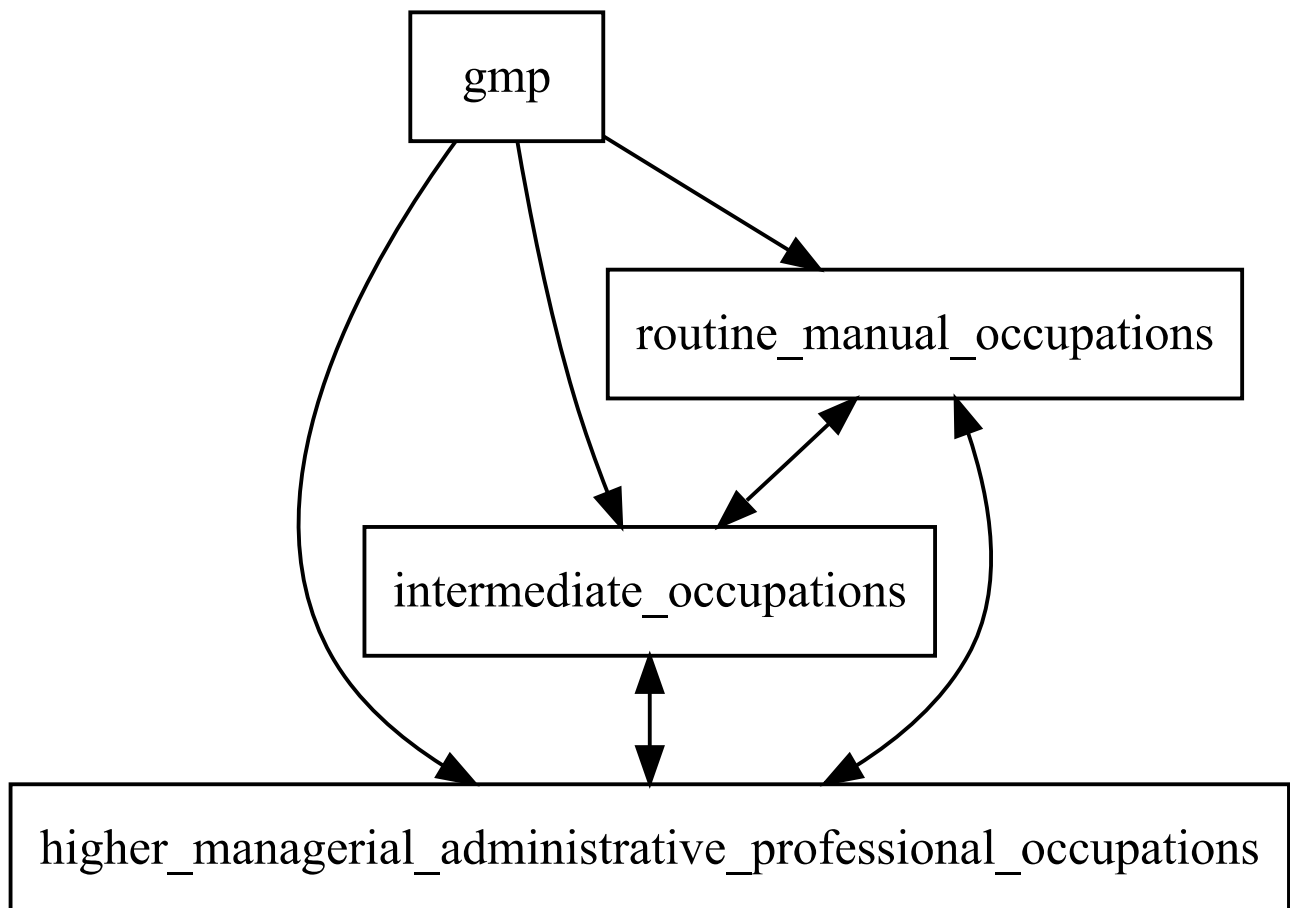
```
-140.858 12.740
```

Laplace approximation to the log-Bayes factor

(experimental; positive values favor object1): -144.929

We can now visualise the results fully by plotting out the full posteriors for coefficient uncertainty and comparing priors with posteriors

```
library(lavaanPlot)
lavaanPlot2(fitbayeswithin,include = "covs")
```



```

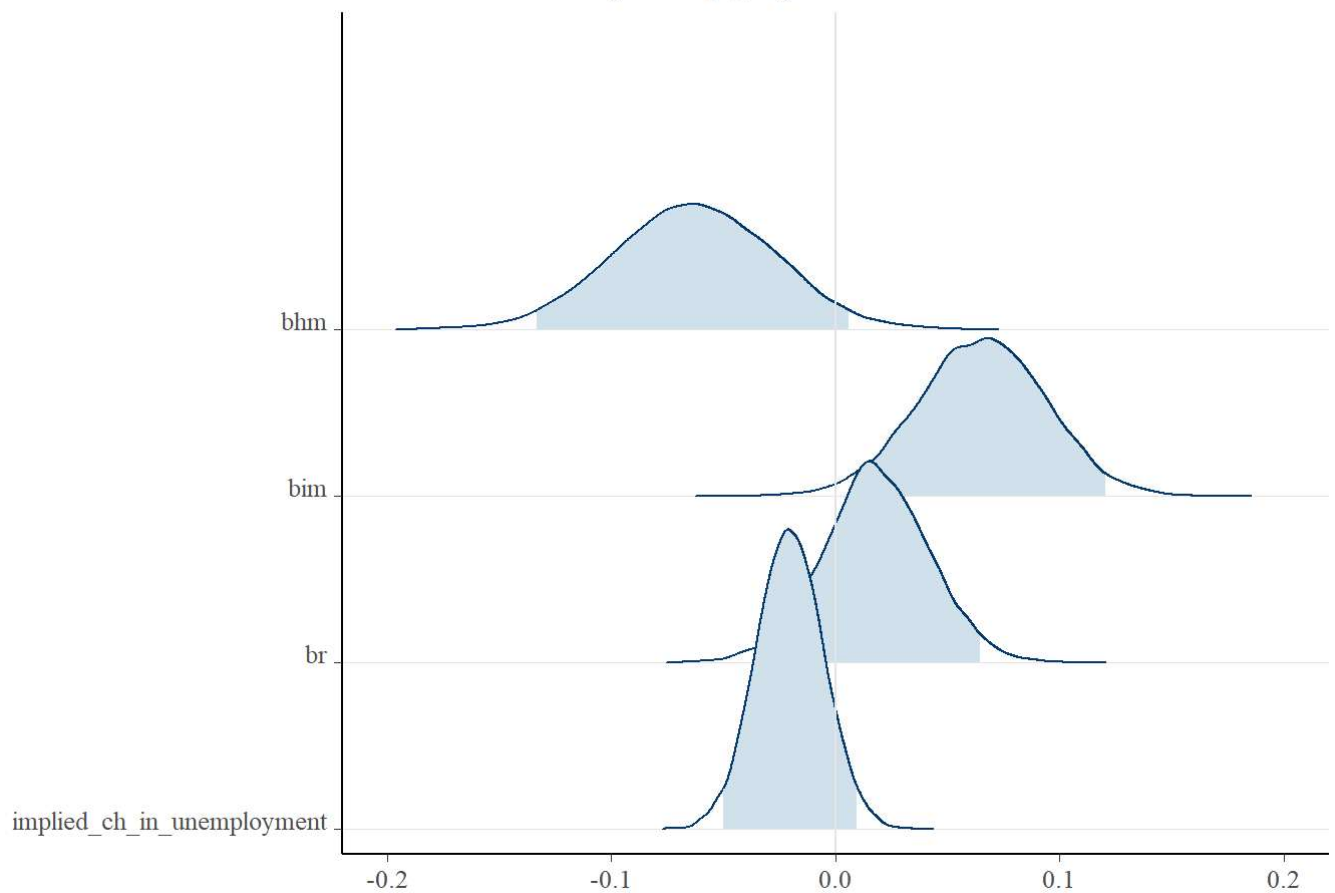
mcmc_areas_ridges(blavInspect(fitbayeswithin,what="draws"),pars = c("bhm","bim","br","implied_
  xlim(c(-0.2, 0.2)) +
  ggtitle("Within Effects of joining gmp across sectors")

```

Scale for x is already present.

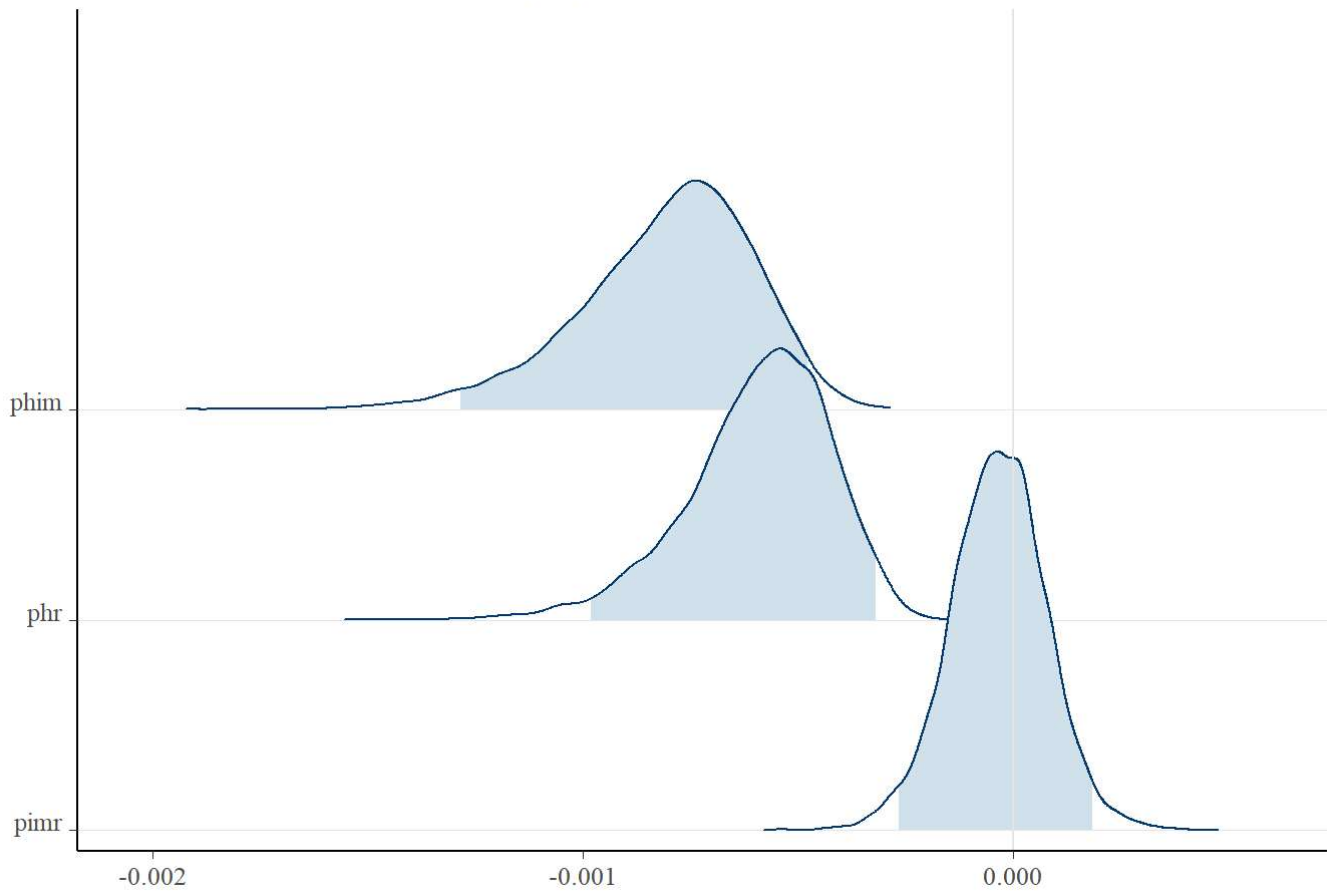
Adding another scale for x, which will replace the existing scale.

Within Effects of joining gmp across sectors



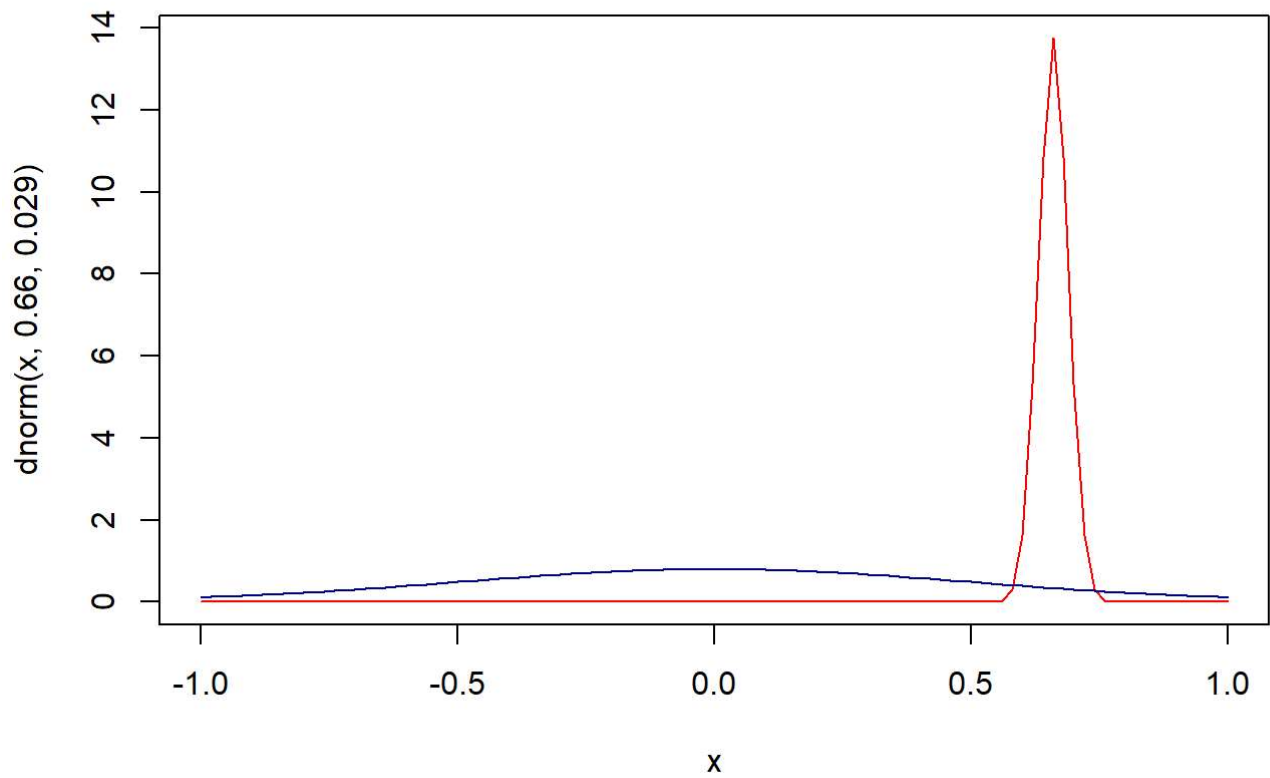
```
mcmc_areas_ridges(blavInspect(fitbayeswithin,what="draws"),pars = c("phim","phr","pimr"),prob  
ggtitle("Within covariences of joining gmp across sectors")
```

Within covariences of joining gmp across sectors



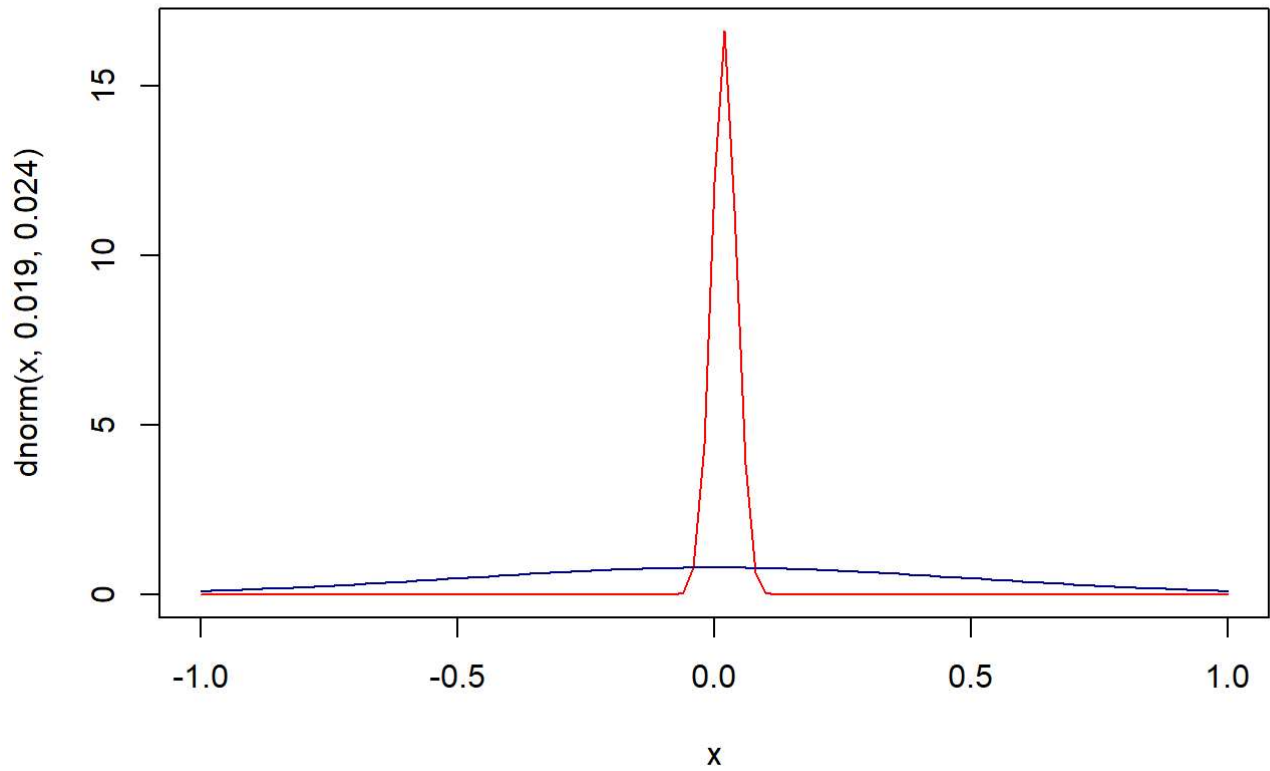
```
curve(dnorm(x,0.66,0.029),from=-1,to=1,col="red",main="  intermediate_occupations prior/poste  
curve(dnorm(x,0,0.5),from=-1,to=1,add=TRUE,col="darkblue")
```

intermediate_occupations prior/posterior



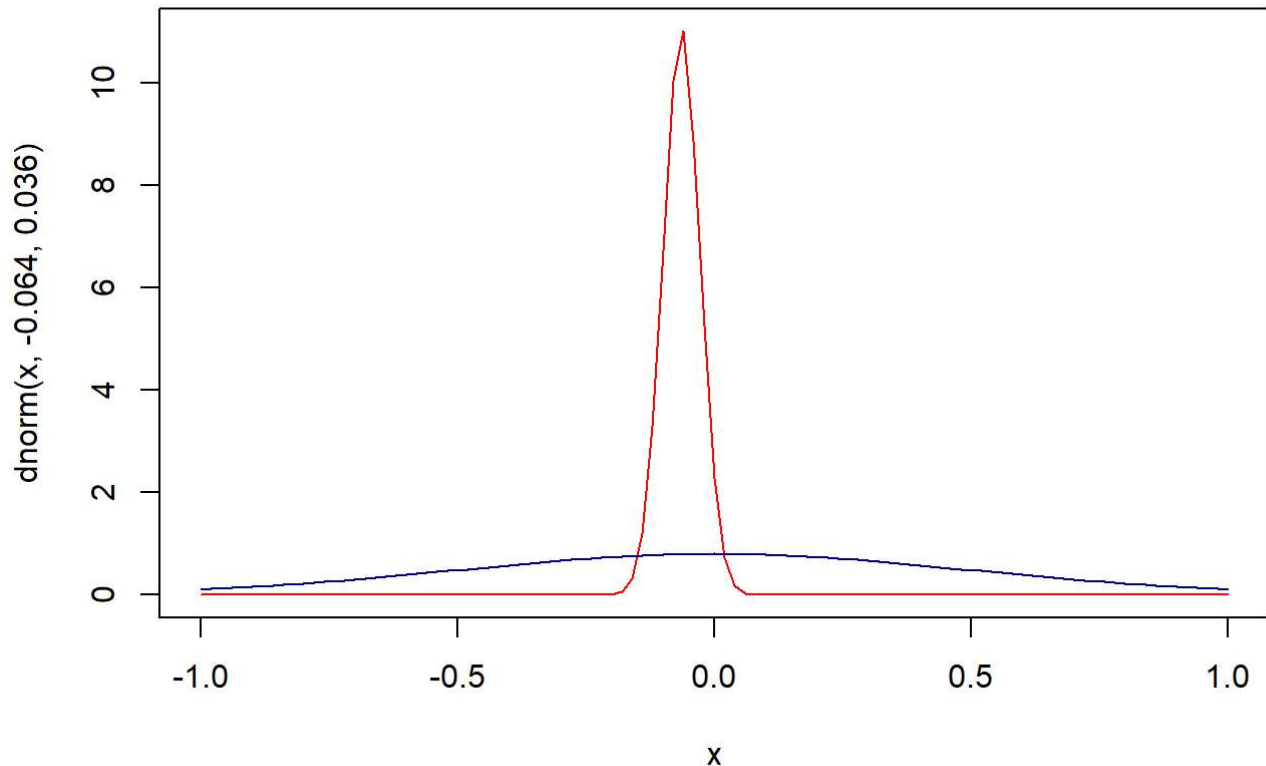
```
curve(dnorm(x,0.019,0.024),from=-1,to=1,col="red",main=" routine_manual_occupations prior/posterior")
curve(dnorm(x,0,0.5),from=-1,to=1,add=TRUE,col="darkblue")
```

routine_manual_occupations prior/posterior



```
curve(dnorm(x, -0.064, 0.036), from=-1, to=1, col="red", main="higher_managerial_administrative prior")
curve(dnorm(x, 0, 0.5), from=-1, to=1, add=TRUE, col="darkblue")
```

higher_managerial_administrative prior/posterior



APPENDIX: Doing it the Classical way. Feel Free to use if don't want to use BAYES

In finite n an alternative to exact inference via MCMC is to use MLE for estimates but bootstrap the standard errors to hopefully attain a simulated sampling distribution which can be close to an asymptotic one empirically. For this particular case the bootstrap attains fairly similar interval estimates compared to the exact bayesian method.

```
library(lavaan)
```

This is lavaan 0.7-2

lavaan is FREE software! Please report any bugs.

```
fitbootpooled = sem(
  sur_model_clean,
  data      = widepanel_clean,
  ncpus     = 12,
  se="bootstrap",
  test = "bootstrap",bootstrap=4000,parallel="snow"
)
```

Warning: lavaan->lav_model_nvcov_bootstrap():
125 bootstrap runs failed or did not converge.

```
summary(fitbootpooled,ci=TRUE)
```

lavaan 0.7-2 ended normally after 107 iterations

Estimator	ML
Optimization method	NLMINB
Number of model parameters	15
Number of observations	54

Model Test User Model:

Test statistic	0.000
Degrees of freedom	0
Test statistic	0.000
Degrees of freedom	0

Parameter Estimates:

Standard errors	Bootstrap
Number of requested bootstrap draws	4000
Number of successful bootstrap draws	3875

Regressions:

	Estimate	Std.Err
higher_managerial_administrative_professional_occupations ~		
gmp (bhm)	-0.039	0.072
estblshd	0.005	0.034
highpeak	0.091	0.051
intermediate_occupations ~		
gmp (bim)	0.019	0.026
estblshd	-0.039	0.013
highpeak	-0.013	0.022
routine_manual_occupations ~		
gmp (br)	0.029	0.052
estblshd	-0.001	0.025
highpeak	-0.073	0.028
z-value P(> z) ci.lower ci.upper		

-0.539	0.590	-0.178	0.109
0.150	0.881	-0.061	0.075
1.803	0.071	-0.009	0.194

0.707	0.480	-0.031	0.072
-3.047	0.002	-0.063	-0.014
-0.607	0.544	-0.055	0.031

0.564	0.573	-0.072	0.133
-0.045	0.964	-0.052	0.047
-2.666	0.008	-0.129	-0.019

Covariances:

	Estimate
.higher_managerial_administrative_professional_occupations ~	
.intrmd_ (phim)	-0.003

```

.rtn_mn_ (phr) -0.007
.intermediate_occupations ~
.rtn_mn_ (pimr) 0.000
Std.Err z-value P(>|z|) ci.lower ci.upper

0.001 -3.305 0.001 -0.005 -0.001
0.001 -5.389 0.000 -0.009 -0.004

0.000 0.484 0.628 -0.001 0.001

```

Variances:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
.hghr_mngrl_d__	0.012	0.002	5.765	0.000	0.008	0.016
.intrmdt_ccptns	0.002	0.000	5.555	0.000	0.001	0.003
.rtn_mnl_ccptns	0.006	0.001	5.678	0.000	0.004	0.008

Defined Parameters:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
gap_routin_mng	0.068	0.119	0.573	0.567	-0.171	0.299
gap_mnl_ntrmdt	0.011	0.057	0.192	0.848	-0.098	0.125
gp_hghm_ntrmdt	-0.057	0.089	-0.646	0.518	-0.225	0.124
polarization	-0.028	0.064	-0.436	0.663	-0.158	0.095
impld_ch_n_nmp	-0.009	0.022	-0.406	0.685	-0.054	0.033

```

fitboot = sem(
  sur_model_within,
  data      = widepanel_demeaned,
  ncpus     = 12,
  se="bootstrap",
  test = "bootstrap",bootstrap=4000,parallel="snow"
)

```

Warning: lavaan->lav_model_nvcov_bootstrap():
359 bootstrap runs failed or did not converge.

Warning: lavaan->lav_model_test():
58 bootstrap runs failed or did not converge.

```
summary(fitboot,ci=TRUE)
```

lavaan 0.7-2 ended normally after 91 iterations

Estimator	ML
Optimization method	NLMINB
Number of model parameters	9
Number of observations	54

Model Test User Model:

Test statistic	0.000
Degrees of freedom	0

Test statistic	0.000
Degrees of freedom	0

Parameter Estimates:

Standard errors	Bootstrap
Number of requested bootstrap draws	4000
Number of successful bootstrap draws	3641

Regressions:

	Estimate	Std.Err
higher_managerial_administrative_professional_occupations ~		
gmp (bhm)	-0.064	0.028
intermediate_occupations ~		
gmp (bim)	0.066	0.027
routine_manual_occupations ~		
gmp (br)	0.019	0.013
z-value P(> z) ci.lower ci.upper		
-2.260 0.024 -0.122 -0.006		
2.448 0.014 0.011 0.117		
1.386 0.166 -0.011 0.044		

Covariances:

	Estimate
.higher_managerial_administrative_professional_occupations ~~	
.intrmd_ (phim)	-0.001
.rtn_mn_ (phr)	-0.001
.intermediate_occupations ~~	
.rtn_mn_ (pimr)	-0.000
Std.Err z-value P(> z) ci.lower ci.upper	
0.000 -3.548 0.000 -0.001 -0.000	
0.000 -3.351 0.001 -0.001 -0.000	
0.000 -0.131 0.896 -0.000 0.000	

Variances:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
.hgmr_mngrl_d__	0.001	0.000	4.152	0.000	0.001	0.002
.intrmdt_ccptns	0.001	0.000	5.225	0.000	0.001	0.001
.rtn_mnl_ccptns	0.001	0.000	3.996	0.000	0.000	0.001

Defined Parameters:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
gap_routin_mng	0.083	0.035	2.359	0.018	0.012	0.157
gap_mnl_ntrmdt	-0.047	0.030	-1.558	0.119	-0.107	0.016
gp_hghm_ntrmdt	-0.130	0.053	-2.442	0.015	-0.235	-0.021
polarization	-0.112	0.053	-2.112	0.035	-0.212	-0.003
impld_ch_n_nmp	-0.020	0.012	-1.710	0.087	-0.044	0.002

